

Prompt Engineering

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Prompt Engineering, also known as In-Context Prompting, refers to methods for communicating with Large Language Models (LLMs) to steer their behavior for desired outcomes without changing the model weights. It is an empirical science and the effect of prompt engineering methods can vary highly among models, thus requiring heavy experimentation and heuristics.

This post focuses on prompt engineering for text-to-text language models. It covers basic techniques, advanced reasoning frameworks, and external steering mechanisms.

Basic Prompting Techniques

Zero-Shot & Few-Shot Learning

Zero-shot prompting feeds the model a direct instruction without any localized examples. The model relies entirely on its pre-trained generalized parameters to execute the completion.

```
Classify the text into neutral, negative or positive.  
Text: I think the vacation is okay.  
Sentiment:
```

Few-shot prompting (Brown et al. 2020) provides a set of high-quality demonstrations showcasing both input and expected structural outputs. This configures the in-context working space to mimic the target formatting directly.

```
Great product, fast delivery! // Positive  
Bad quality, returned it immediately. // Negative  
It works as described, nothing special. // Neutral  
The dynamic response was completely flawed. //
```

Advanced Reasoning Frameworks

Chain-of-Thought (CoT)

Introduced by **Wei et al. (2022)**, Chain-of-Thought prompting instructs the model to produce intermediate reasoning steps before delivering the final answer. This is highly effective for

complex arithmetic, symbolic manipulation, and multi-step logic puzzles. It can be triggered simply by appending sentences like *"Let's think step by step"* (Kojima et al. 2022).

Self-Consistency Sampling

Wang et al. (2022) proposed Self-Consistency reasoning. Instead of taking a single greedy decoding path, the model samples multiple diverse reasoning paths (using a non-zero temperature configuration). The final answer is selected via a majority vote among the generated outputs. This significantly decreases random generation hallucinations in mathematical domains.

Tree of Thoughts (ToT)

Yao et al. (2023) extended CoT into the Tree of Thoughts framework. ToT structures the reasoning space as a tree where each node represents a coherent "thought" or intermediate step. It enables models to explore multiple paths, evaluate its own current progress via self-prompting classifiers, and use classic search algorithms like Breadth-First Search (BFS) or Depth-First Search (DFS) to look ahead or backtrack when hitting dead ends.

Automatic Prompt Design

Manually constructing optimal prompts is highly time-consuming and fragile. Automated frameworks treat prompt generation as an optimization problem over discrete text spaces.

- **APE (Automatic Prompt Engineer; Zhou et al. 2022):** Uses an LLM to generate candidate prompt instructions based on input-output pairs, then evaluates the choices using an execution score to optimize the instructions iteratively.
- **CONN (Chain-of-Notion):** Enhances stability by restructuring unstructured knowledge sources into formalized internal premises before formulating instructions.

External Control & Structural Injection

Directional Stimulus Prompting

Li et al. (2023) introduced a framework where a small, tunable policy model generates a "hint" or dynamic stimulus keyword for each specific input. This hint is injected into the prompt to guide a frozen LLM toward the desired target text properties without requiring massive foundational updates.

ReAct (Reasoning and Acting)

Developed by **Yao et al. (2023)**, ReAct forces a tight integration between linguistic reasoning and discrete action loops. The model follows an iterative cycle: **Thought** → **Action** → **Observation**.

Question: What is the elevation peak of the mountain where the film 'Inception' shot its winter fortress scenes?

Thought: I need to find the location where the winter fortress scenes were shot, then lookup its elevation.

Action: Search[Inception winter fortress filming location]

Observation: The fortress scenes in Inception were filmed at the defunct Fortress Mountain resort in Alberta, Canada.

Thought: I now need to search for the elevation peak of Fortress Mountain.

Action: Search[Fortress Mountain Alberta elevation]

Observation: Fortress Mountain has an elevation peak of 2,753 meters (9,032 feet).

Thought: I have the final answer.

Answer: 2,753 meters

Prompting Best Practices & Heuristics

Strategy	Core Practice	Expected Impact
Clear Delimiters	Wrap distinct inputs inside triple quotes, XML tags (e.g., <code><text></code>), or markdown blocks.	Prevents prompt injection attacks and parsing confusion.
Specify Output Structure	Explicitly demand structural signatures like JSON schema, tables, or precise bullet counts.	Improves compatibility with external database parsers and downstream code.
Negative Constraints	Provide explicit rules of what <i>*not*</i> to do (e.g., "Do not use technical jargon").	Reduces stylistic deviation and limits off-topic text expansions.
Example Ordering	Arrange few-shot examples carefully; place complex or highly representative cases closer to the end.	Combats recency bias inherent in long sequence attention maps.

Core Vulnerabilities & Security Pitfalls

1. **Prompt Injection:** Attackers craft user inputs to hijack the model's core instruction set, overriding safety rules or system prompts.
2. **Prompt Leaking:** Strategic questioning tricks the model into outputting its hidden initial system configuration instructions.

3. **Goal Hijacking:** Forcing an agent to discard its intended specialized business logic function to execute unauthorized tasks (e.g., writing poems instead of calculating financial routing values).